

The Lyndon B Johnson Problem: Computational Findings Regarding the “Friendly Seating Problem”

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A School Which Contains Bathrooms

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Abstract: This paper models one-dimensional optimization in social settings where individuals sequentially choose their position, focusing on the first player, or LBJ, who seeks proximity to others rather than avoiding it. Building on existing models such as “The Urinal Problem”, we investigate scenarios where such a position can impact future utility. The findings offer new insights into the placement strategies of the "First Player" or LBJ. (see Appendix)

1. Introduction

Combinatorial, Operations, and Game Theoretic research has long explored optimizations within sequential movements of individuals. Particularly, in social gatherings, the simple act of choosing one's position and proximity to others can significantly influence the dynamics of interactions [1]. In such social and networking settings, such as conferences or bathroom urinals, it is therefore valuable to understand potential optimizations contingent on the expected behavior of others. In particular, the most defining variable in such an arrangement — the first player.

Present research, such as the “unfriendly seating arrangement” [2] and “unfriendly theater seating arrangement problem” [3] regarding the modeled optimizations of such has primarily focused on and assumed the interaction-averse nature prevalent in the average computer scientist and mathematician. Recent research has focused on an anti-proximity urinal

interpretation of the one-dimensional optimization [4] [6]. However, circumstances may present where the first player may desire such interaction or proximity, such as the explored Lyndon B Johnson urinal bijection in the appendix — the namesake of this paper.

The solutions to friendly problems may be further applied to scenarios in which future utility may be affected by the positioning of the first player and their proximity to succeeding players. Examples of such could be found in Business Analytics, such as in the placement optimization of a firm positively affected by the proximity of potential succedents, succedents which may be aversive to said proximity.

Thus, in this paper, we will explore one-dimensional optimizations of a proximity-seeking first player, henceforth also referred to as LBJ (referring both to Lyndon B. Johnson, the namesake of our problem as explained in the Appendix, and to our goat, LeBron James).

The particular models will explore a "Friendly Seating Problem" game with 10 positions, or urinals, and ten players. Each player will enter sequentially at a distinct, discrete time (t ranging from initial position 0 to 10). From here, we explore and optimize three differing models of the utility function for LBJ.

2. Models

For each following models, we assume all succeeding players after the first player are aversive to interaction, choosing a random vacant position not currently adjacent to an occupied position. If all of such positions are taken, they will choose a random vacant position. In each, the utility of the first player is defined differently, thus influencing optimal placements. For each model, 100,000 simulations were performed, with the data averaged and presented.

The Privacy Model

In this problem, the first player intends to optimize for the least amount of privacy, defined by the distance between the first player and the closest adjacent player. Thus, we average the privacy across the times $t=2$ to $t=10$.

Firstly, we calculate privacy as the simple distance from the first player to the closest player at each time t . The average privacies are as follows:

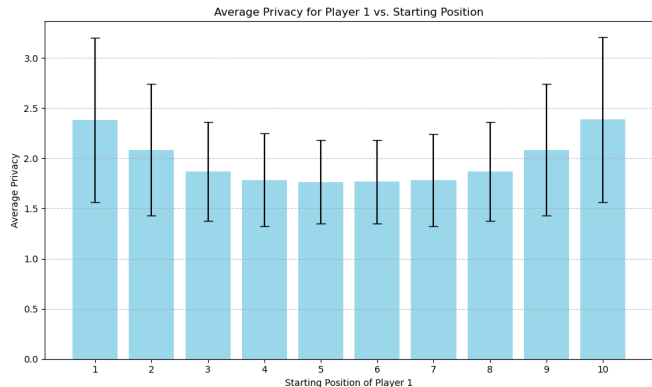
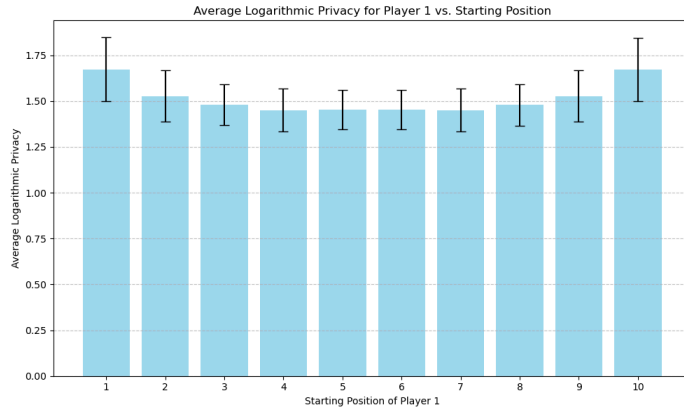


Figure 1: Simulation Results of The Privacy Model



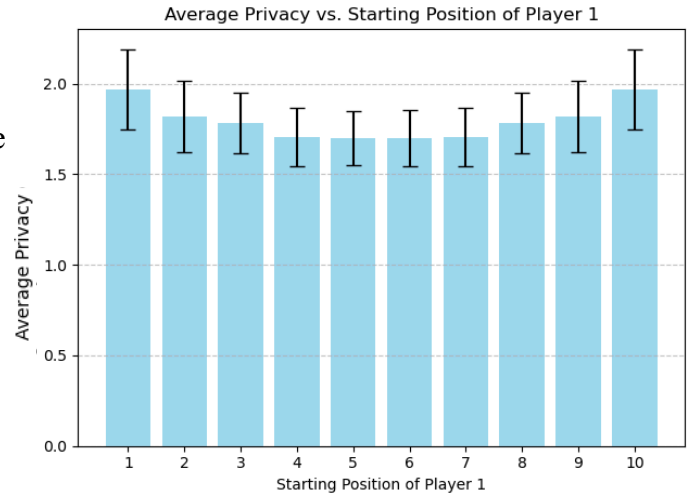
Next, we consider privacy logarithmically, in which any increase in distance beyond 1 is seen as less significant. The formula for privacy is thus given by $1 + \log(x)$, in which x is the distance to the closest player.

Figure 2: Simulation Results of The Logarithmic Privacy Model

From these models, we conclude the optimal position that the first player, in which they experience the least privacy by both privacy calculations, is given by a position in the center of the one-dimensional line of positions. However, interestingly, the difference between center positions is relatively insignificant.

The Malicious Model

In this model, as opposed to solely optimizing for their lack of privacy, the first player hopes to decrease the average privacy of the remainder of the group. This privacy is given by the simple distance, calculating the average of each player across the times when they are in a position. Then, the model computes the average across players 2 to 10.



Here, we once again conclude an optimal placement in the middle of the positions.

Figure 3: Simulation Results of The Malicious Model

The Boolean Model

This model treats proximity as a boolean — either the first player is proximal enough or not. Here, we analyze across differing “intervals” of proximity, where an interval of x suggests another player x positions away is proximal enough. In particular, we see below data from ranges 1, 2, and 3, showing how long in time t until a player becomes proximal enough.

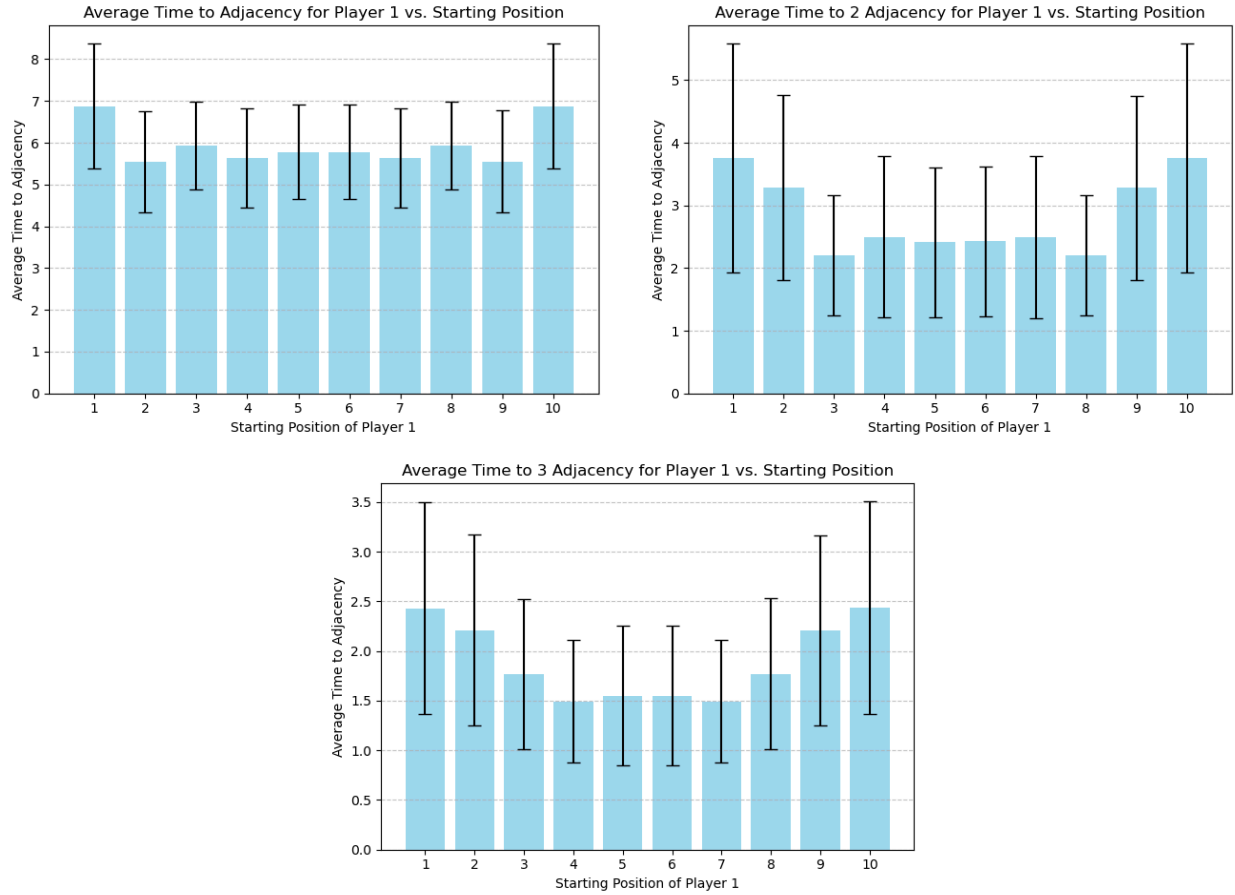


Figure 4: Simulation Results of The Boolean Model

From such, we conclude a compelling empirical finding:

The Privacy Interval Symmetry Strategy Conjecture: When optimizing for the shortest time of x -range privacy, the optimal position among a one-dimensional group of positions is always x positions from each end.

3. Conclusion

From our first three-player utility models, we conclude a general trend of center positions presenting as optimal. Interestingly, the Privacy Interval Symmetry Strategy Conjecture demonstrates an interesting positioning for the specific Boolean case.

Moving forward, many extensions and explorations regarding said "Friendly Seating Problem" are yet to be explored. In particular, our models assumed all succeeding players to choose

randomly among vacant positions not adjacent to those taken, then move to any random vacant position. Differing selection algorithms may prove interesting in further exploration of this problem, such as those presented in [4]. All conclusions from this paper are also empirical, and a rigorization may prove valuable.

Higher-dimensional analysis may also prove to be interesting. In particular, two-dimensional analysis may apply to management and marketing issues pertaining to proximity interactions between players of differing motives. Most strictly, the conclusions of this paper align closely with the case explored within the Appendix, aligning with current research [4] [6].

4. Appendix

A direct bijection of our problem is presented in the Lyndon B Johnson Urinal Problem. Specifically, Lyndon B Johnson was rumored to have certain tendencies, specifically towards a secret service agent, which may incline him towards utility in urinal proximity [7]. Under said interpretation, our initial Privacy and Malicious Models may prove somewhat relevant for optimising his proximity to potential targets. However, our third Boolean Model serves most directly, where the interval represents LBJ's effective range, thus adjusting his initial position in accordance with the Privacy Interval Symmetry Strategy (PISS) Conjecture. Thus, this paper serves at its core to justify and investigate optimizations for LBJ's Urinal problem, seeking proximity to others and applying PISS.

5. Citations

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